Collocation Dynamics in Physics-Informed Neural Networks

Adaptive sampling, diffusion, and solution sharpness

Hiero de Paula, Sofia Suesdek

Escola Politécnica da Universidade de São Paulo

October 20, 2026

Why study collocation dynamics?

The physics loss is evaluated on a finite set of points:

$$\mathcal{L}_{\mathrm{PDE}}(\theta; Z) = \frac{1}{N} \sum_{i=1}^{N} r_\theta(z_i)^2, \qquad z_i = (x_i, t_i).$$

- Point locations determine where the PDE is enforced during training.
- Keeping, replacing, or moving points changes coverage and the optimization trajectory, even at the same point budget.
- Concentrating points near difficult features may help, while leaving other regions insufficiently constrained.

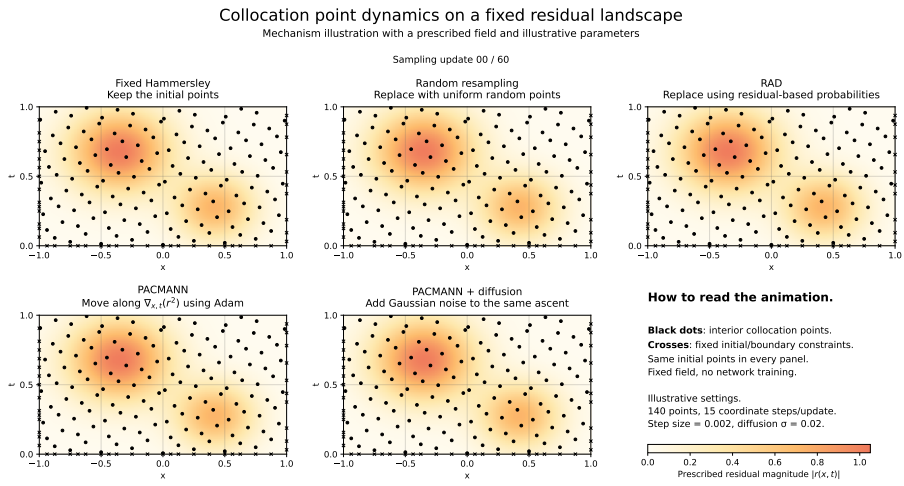
Five ways to place the same interior point budget

Method	Interior point strategy
Fixed Hammersley	Keep the initial low-discrepancy design.
Random resampling	Periodically replace points with fresh uniform random points.
RAD[1]	Sample fresh candidates preferentially where residual magnitude is large.
PACMANN[2]	Move points along the coordinate gradient of squared residual, using Adam ascent.
PACMANN + diffusion	Add domain-scaled Gaussian perturbations to the coordinate-Adam motion.

Residual-based selection versus gradient-based adaptive sampling

RAD selects a new population using residual scores; PACMANN moves the current population using $\nabla_{(x,t)} r_\theta^2$.

Collocation sampling: a mechanism illustration



Research questions

1. How does the benefit of point-adaptive collocation change across solution-sharpness regimes, for solution and PDE-residual error?
2. Does adding diffusion to PACMANN improve either outcome, and does the effect depend on the regime?

Working hypothesis

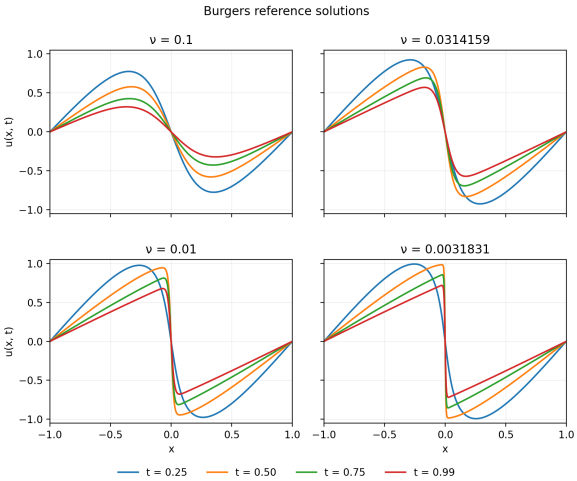
Gradient-adaptive methods may offer greater benefits for solutions with sharp features. Both the adaptive-sampling and diffusion benefits remain open questions.

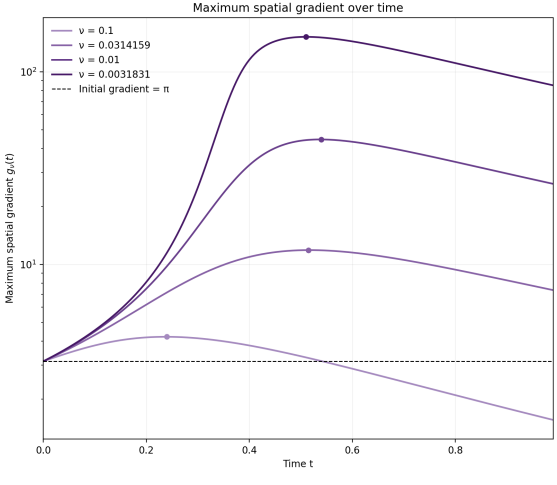
Starting problem: viscous Burgers equation

$$\partial_t u + u\,\partial_x u - \nu\,\partial_{xx} u = 0, \qquad (x,t) \in [-1,1] \times [0, 0.99].$$
$$u(x,0) = -\sin(\pi x), \qquad u(-1,t) = u(1,t) = 0.$$

- Viscosity ν controls diffusion in the PDE and the solution regime.
- Train from the PDE and fixed initial/boundary constraints.
- Use numerical reference solutions only for evaluation.

Reference solutions across viscosity regimes





Colors denote time (left) and viscosity (right). $g_\nu(t) = \max_x |\partial_x u_\nu(x,t)|$.

Released PACMANN code to a controlled study

Category	Implementation choice
Retained	Four 64-neuron tanh layers, float64, five alternating Adam/L-BFGS stages, sampler settings, coordinate-Adam procedure, keep initial/boundary constraints fixed.
Changed	Match 2,500 interior PDE points from initialization for every method; Use generated references and common final-checkpoint evaluation.
Added / tuned	Gaussian diffusion of point coordinates; select its coefficient separately at each viscosity on tuning seeds.

- DeepXDE with a PyTorch backend, adapted from the released implementation.
- Evaluate final models and retain all planned seeds, including failures.
- Record actual optimizer work; shared limits do not imply equal computation.

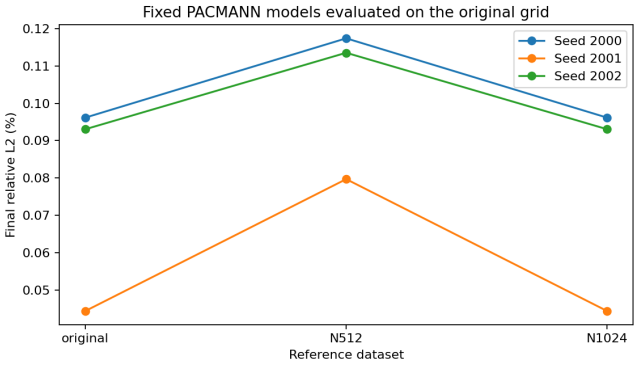
Reference generation and validation at $\nu = 0.01/\pi$

Generated reference

- Chebyshev spatial collocation, degree $N = 1024$.
- Radau time integration.
- Compare on the original 256 spatial points and 100 snapshot times.

Two different checks

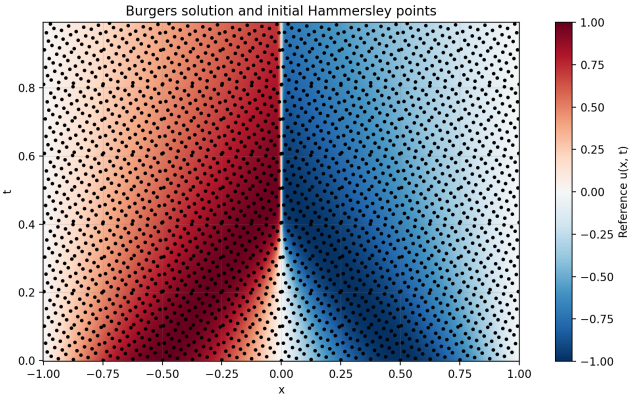
- Reference disagreement: 0.000319% relative L^2.
- N1024 change in fixed-model error scores: less than 0.006% relative.



Each line is one fixed trained model. Only the reference used to score its predictions changes; no retraining occurs.

Generated values are barycentrically interpolated in space at exact shared times. Agreement supports N1024 at this viscosity; it does not certify other viscosities or reproduce published training results.

Matched training and independent evaluation



Training controls

- Paired seeds and initial weights.
- 2,500 PDE; 80 BC; 160 IC.
- Interior updates during Adam only.

Final-model evaluation

- Solution: uniform 1024×199 grid.
- Residual RMS: independent 1024×200 midpoint grid.
- Boundary / initial RMSE.

Initial Hammersley points at $\nu = 0.01/\pi$; paired across methods.

Float64; DeepXDE 1.14.0 / PyTorch 2.7.1; RTX 5070 (12 GB). Ten concurrent seeds per method; method batches run sequentially.

Tune first, freeze settings, then evaluate

Development: three paired seeds

- Seeds 3000–3002.
- Test $\sigma = 0$ and positive values $10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}$.
- Select by mean final solution relative L^2 error.

Evaluation: ten paired seeds

- Freeze sigma before using seeds 4000–4009.
- Use the same sigma for residual-error evaluation.
- Compare all five methods; retain every planned seed.

Frozen diffusion settings: $\nu = 0.01/\pi \Rightarrow \sigma = 10^{-4}$; $\nu = 0.01 \Rightarrow \sigma = 10^{-3}$.

At $\nu = 0.01$, zero is a comparison baseline only: select the best positive sigma and expand a boundary winner under the approved stopping rule. The earlier sweep allowed zero as a candidate. Other samplers retain upstream settings; disclose this tuning asymmetry.

Results: solution and residual accuracy

Results to be added

Complete this comparison after the $\nu = 0.01$ campaign is finished and verified. Include the accepted $\nu = 0.01/\pi$ campaign and the new campaign.

- TODO: Compare all five methods for both error metrics at each viscosity.
- TODO: Show ten-seed means and sample standard deviations; label units.
- TODO: Link each table or figure to its frozen campaign and configuration.

Results: variability and paired comparisons

Analysis to be added

Use the complete seed sets, including unfavorable and failed outcomes.

- TODO: Show individual-seed distributions alongside aggregate results.
- TODO: Assess paired diffusion-minus-PACMANN changes for both metrics.
- TODO: Compare adaptive methods with fixed Hammersley and random resampling.
- TODO: Describe large-error outcomes and the sensitivity of means; keep any trimmed means supplementary.

Analysis: what changes across viscosities?

Interpretation to be added

Connect verified campaign comparisons to the two research questions.

- TODO: Assess whether the effect of diffusion changes across viscosities.
- TODO: Compare method benefits for solution and residual error separately.
- TODO: Discuss variability, reference accuracy, and per-viscosity tuning.

A viscosity-dependent ranking alone does not establish a sharpness effect. Sharpness and feature-region definitions must be independent of adaptive sampling.

Conclusions, limitations, and next steps

- **Evidence:** TODO: State what the completed comparisons support about adaptive collocation and diffusion.
- **Limitations:** TODO: Summarize reference accuracy, seed variability, and the scope of the tested viscosities.
- **Next step:** TODO: State the next bounded study decision.

Interpretation boundary

Distinguish observations from hypotheses about their causes. Improvement in solution error need not imply improvement in independent residual error.

References

[1] C. Wu, M. Zhu, Q. Tan, Y. Kartha, and L. Lu.
A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks.
2023.

[2] C. Visser, A. Heinlein, and B. Giovanardi.
PACMANN: Point adaptive collocation method for artificial neural networks.
2025.

Released-code provenance and controlled-study deviations are recorded in `references/upstream.md`.

Backup: detailed scientific settings

Setting	Frozen configuration
Network / precision	2–64–64–64–64–1, tanh, Glorot normal; float64.
Point budgets	2,500 interior PDE points; 80 boundary and 160 initial points.
Optimizer schedule	Five phases, each with 7,000 Adam steps at 10^{-3}, followed by L-BFGS.
L-BFGS limits	Nominal maxiter 3,000, maxfun 3,750; maxcor 100, gtol 10^{-8}, ftol 0.
Sampling cadence	Every 50 Adam steps; no point updates during L-BFGS.
PACMANN motion	15 coordinate steps at 10^{-5}; $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$ outside the square root.
RAD	100,000 candidates; weight $\lvert r\rvert / \operatorname{mean}(\lvert r\rvert) + 1$; sample without replacement.
Diffusion convention	Per coordinate step, add independent Gaussian noise with componentwise standard deviation $\sigma(L_x, L_t)/\sqrt{15}$.
Evaluation	Uniform solution grid 1024×199 with spatial barycentric reference interpolation; independent residual grid 1024×200.

The retained DeepXDE L-BFGS driver can exceed nominal stage limits. Report recorded iterations, function evaluations, and training time.